Composites: Part A 191 (2025) 108720

Contents lists available at ScienceDirect

Composites Part A

journal homepage: www.elsevier.com/locate/compositesa

# Mechanical role of graphene nanofiller on vibration damping properties of highly cross-linked polymers

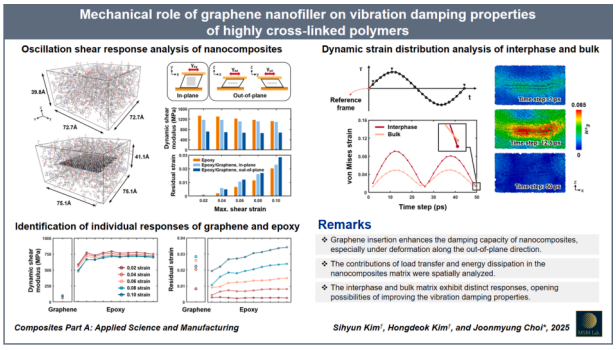
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## HIGHLIGHTS

- Nanomechanical damping properties of graphene-functionalized epoxy was investigated.
- The 2D structure of graphene leads to strong direction-dependent damping behavior.
- Load transfer and energy dissipation of local matrix region was spatially analyzed.
- Spatial-dependent response is due to structural change by heterogeneous interaction.
- Molecular-level interphase design strategy to improve vibration damping is presented.

## GRAPHICAL ABSTRACT




## ARTICLE INFO

*Keywords:*
Polymer nanocomposites
Vibrational damping
Interphase
Heterogeneous filler
Molecular dynamics

## ABSTRACT

This study investigates the effect of graphene incorporation into epoxy on its microstructure and energy dissipation properties under vibration loading. Using all-atom molecular dynamics simulations, the load-carrying and energy dissipation capabilities of graphene/epoxy nanocomposite models were successfully quantified at the nanoscale. Oscillatory shear deformation analysis revealed that graphene enhances the interfacial energy dissipation under out-of-plane loading, which significantly improves the damping performance of the nanocomposites. However, a detailed spatial analysis of the matrix domain revealed that the interphase around graphene is inferior to that of the bulk matrix in both load-carrying and energy-dissipation capabilities, which is attributed to the locally elevated cross-linking density at the interphase and insufficient interfacial interactions. This encapsulated interphase structure causes a simultaneous increase in deformability and elasticity under external shear strain. The study findings underscore the necessity of optimizing the interfacial structure for developing graphene/epoxy nanocomposites with superior passive vibration damping properties.

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https://doi.org/10.1016/j.compositesa.2025.108720
Received 20 November 2024; Received in revised form 27 December 2024; Accepted 9 January 2025
Available online 11 January 2025
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## 1. Introduction

Vibration damping control is an industrial problem that has been gaining importance in both traditional machinery sectors and modern high-precision industries for more than 20 years [1–3]. The lack of damping in aircraft wings causes structurally transmitted vibration instabilities during flight, which reduces passenger comfort [4]. For automotive brake pads, damping treatment is essential to minimize vibrations and acoustic noise in passenger compartments [5]. Controlling vibrations during drilling or milling machinery operations directly improves the process performance, dimensional accuracy, and tool life [6]. In gas turbine blades, vibration fatigue failures occur over long periods of use, which account for up to 42 % of all failures [7]. Vibrations in wafer handling and transport systems used in semiconductor processing can cause wafers to shift or rattle, resulting in less stable processing conditions and reduced equipment efficiency [8].

The most fundamental and effective way to maximize the damping capacity of a structure is to intelligently design materials constituting the system to achieve optimal performance. A passive damping design in which energy is disseminated through the internal structure of the material is economical—it does not require an external energy supply to reduce system vibration, which significantly reduces the size and weight of the entire system [9,10]. Recent studies have reported composite materials containing nano-sized particles that can effectively dissipate small-amplitude energy at high frequencies [11–13]. The damping performance of such nanocomposites mainly originates from the molecular-level friction at the filler-matrix interface [14]. The interphase shaped by heterogeneous systems forms an imperfect bond owing to differences in mechanical properties, such as stiffness. The repeated separation and reattachment of the interface due to external deformations causes energy to dissipate as frictional heat. The kinematic characteristics because of such imperfect bonding become more evident as the difference in properties between the filler and matrix increases. For example, epoxy composites containing nano-sized fillers have extremely high surface areas occupied by such interfaces. Therefore, the vibrational energy for a desired bandwidth can be effectively dissipated by designing appropriate interphase conditions.

A promising filler that achieves passive damping in polymer nanocomposites is the nanocarbon series. Carbon nanomaterials, such as carbon nanofibers, carbon nanotubes, and graphene, are well known for their low density, high fracture strength, and elastic modulus [15,16]. These properties lead to high damping performance and a wide range of vibration absorption capabilities, making polymer nanocomposites excellent for vibration reduction [17–19]. Among these, graphene, which has a two-dimensional planar structure, has an extremely large contact area with the substrate in the interfacial region and is effective in dissipating frictional energy [20]. Katsiropoulos et al. demonstrated that the incorporation of graphene enhances the damping ratio of both pristine polymers and carbon fiber/polymer composites [21]. Sujon et al. ascribed this enhancement to the interfacial “stick-slip” behavior [22]. Pan et al. performed comparative experiments on nanocomposites using carbon nanotubes and graphene as reinforcements in epoxy. The insertion of a small amount of graphene improved the damping ratio of the material by 128.9 %, whereas the same amount of carbon nanotubes amplified it by only up to 41.1 % [23].

Although the above mentioned studies clearly demonstrate the effectiveness of introducing graphene for passive vibration damping of polymer matrices, relatively few studies have quantified the load transfer and energy dissipation properties of the interphase from a theoretical perspective. This understanding is particularly important when nanoscale fillers are inserted into polymer systems, because they can dramatically change the microstructure and physical properties of the surrounding regions. In cross-linked epoxy networks, when the filler is embedded in a polymeric matrix, a highly adsorbed polymer layer is generated in the immediate vicinity of the filler surface [24]. In addition, a physical gap on the angstrom scale is formed because of the inherent non-covalent equilibrium distance between the filler and matrix [25,26]. The locally heterogeneous cross-linking density of the interphase has a significant impact on the mechanical behavior and load-bearing properties of the microstructure [27]. Moreover, the network formed wide discontinuous planes along the insertion direction of graphene because the inserted fillers acted as spatial obstacles that hindered the formation of continuous cross-links. In summary, the change in the epoxy morphology owing to graphene insertion and its effect on the cross-linking properties have a significant impact on the response to oscillatory shear strain. Therefore, to enable an effective theoretical design, both the structural properties formed during the fabrication stage of nanocomposites and mechanical behavior characteristics of real operating environments must be analyzed in the nanoscale domain.

Motivated by the above, in this study, the spatial analysis of the energy dissipation contribution of graphene-functionalized epoxy nanocomposites was theoretically explored in the nanoscale regime. The mechanical behavior of each component and spatial location in the epoxy nanocomposites were quantitatively understood using an all-atom molecular dynamics (MD) simulation solver and a subcontinuum analysis technique. In particular, the damping and energy-dissipation characteristics of the microstructure under an externally applied oscillatory shear strain in a specific direction were quantitatively evaluated. This approach allowed us to theoretically validate that the mechanical response of the epoxy network in the region surrounding graphene is clearly distinct from that in the bulk region. Thus, the morphological transformation of the interphase due to graphene insertion and resulting changes in the cross-linking density of the epoxy network were demonstrated, and the energy dissipation performance exhibited by the unique interphase structure thus formed was quantified. Therefore, this study provides not only important molecular-level insights into the energy dissipation mechanism of nanocomposites but also a theoretical basis for optimizing their performance by improving their microstructure.

## 2. Models and methods

### 2.1. Construction of all-atom models of epoxy and epoxy/graphene nanocomposites

All-atom modeling of the cross-linked epoxy and epoxy/graphene nanocomposites was performed using a commercial MD software, Materials Studio 2020 (Dassault Systèmes, France). Static and dynamic mechanical tests were performed using an open-source MD code, LAMMPS (Sandia Labs, USA). All interactions between the constituent atoms were described using a polymer consistent force field (PCFF) [28].

The diglycidyl ether of bisphenol A (DGEBA) and diethyl toluene diamine (DETDA) were used as the epoxy resin and hardener, respectively. Initially, 300 resins and 150 hardeners were randomly distributed within a periodic unit cell. To prevent unintended interference between molecules during initial positioning, the unit cells were prepared at a very low density of 0.1 g/cc. To densify the prepared model, the length of the unit cell was reduced equally in all directions until it reached 1.0 g/cc. Energy minimization was performed using the conjugate gradient method, with an energy cutoff of 0.001 kcal/mol for every 0.1 g/cc increase in density. After the system density reached 1.0 g/cc, an NVT ensemble for 1 ns at 300 K followed by an NPT ensemble for 1 ns at 300 K and 1 atm were performed.

An epoxy model with a network structure was constructed by simulating the crosslinking reaction between the resin and hardener molecules. The reactive sites of the resin and hardener were identified, and the distances between these sites were continuously monitored throughout the simulation. The cross-linking reaction is determined by the formation of new covalent bonds when the measured distance between the reactive sites is shorter than the cutoff distance. The cutoff value was initially set at 3 Å and gradually increased by 0.5 Å until the

final cross-linking ratio of the network reached 80 %. Energy minimization using the conjugate gradient method was applied after the cross-linking reaction was completed. To allow the microstructure to reach a thermodynamic equilibrium, an NVT ensemble at 300 K for 2 ns and an NPT ensemble at 300 K and 1 atm for 2 ns were applied sequentially. Finally, the model was annealed at a high temperature of 550 K for 2 ns and slowly cooled to 300 K for 4 ns. The epoxy/graphene nanocomposite was prepared using the same process as that used for the epoxy polymer. The only difference between the two models is that a single-layer graphene nano-sheet of 5.2 × 5.1 $nm^2$ is inserted in the epoxy/graphene nanocomposites. The weight fraction of graphene was set to 8.63 wt%, slightly exceeding the typical experimental range, to allow a focused analysis of the interphase. The simulation box was subjected to periodic boundary conditions in all directions, ensuring that the inserted single graphene sheets were distributed uniformly across the model, thus eliminating potential artifacts such as agglomeration. This configuration enables the nanocomposite model to specifically elucidate the interactions between the matrix and the filler in the interfacial region. The final equilibrated configurations of the epoxy and epoxy/graphene models are shown in Fig. 1**(a)**.

Even after the cross-linking reaction simulation was completed, unreacted sites inevitably remained in both the resin and hardener, and the bonding states of the molecules were classified accordingly. Specifically, the epoxy components are classified as free, dangling, and cross-linked based on the number of cross-links, whereas the hardeners are further categorized as primary, secondary, and tertiary according to the amine type, as shown in Fig. 1**(b)** [29,30]. By calculating the type of chemical structure and weight fraction occupied within the microstructure, it was possible to verify the validity of the cross-linked network topology formed from the simulations. The results of topological analysis of the network formed by the considered models are presented in Fig. 1**(c)**. A high cross-linking ratio of 80 % indicates that the network predominantly consisted of cross-linked chains, followed by dangling and free chains. Tertiary amines were the most prevalent in the cross-linked structures. In particular, the proportions of each type of molecule are in good agreement with other observations for epoxy systems with a cross-linking ratio of approximately 80 %, confirming the validity of the simulation methodology used in this study [31–33]. Interestingly, the epoxy/graphene nanocomposite model exhibits topological properties similar to those of the neat epoxy model. This shows that the epoxy network that constitutes the nanocomposite model also has topological properties similar to those of the bulk material. Therefore, the differences in the properties between the neat epoxy model and epoxy/graphene nanocomposite model can be attributed to the presence of graphene.

Note that the distance-based crosslinking used in this study does not account for reaction details, including activation energy or the steric configuration of molecular components [34–36]. The method is particularly suitable for forming appropriate network morphologies in MD simulation environments that can only cover a limited time scale. Even without considering the movement and floating of molecular groups involved in the actual network formation process, the topology of the network structure generated by the considered method aligns well with previous studies (Fig. 1**(c)**). Moreover, the mechanical properties of the neat epoxy were largely consistent with experimental results, confirming that structural distortion of the modeled network was effectively excluded. Based on the predicted topological characteristics and mechanical properties, the crosslinking simulations performed in this study are considered suitable for constructing an epoxy matrix model.

### 2.2. *Mechanical properties characterization*

The mechanical properties of the epoxy and epoxy/graphene nanocomposites were evaluated using uniaxial tensile and simple shear tests.

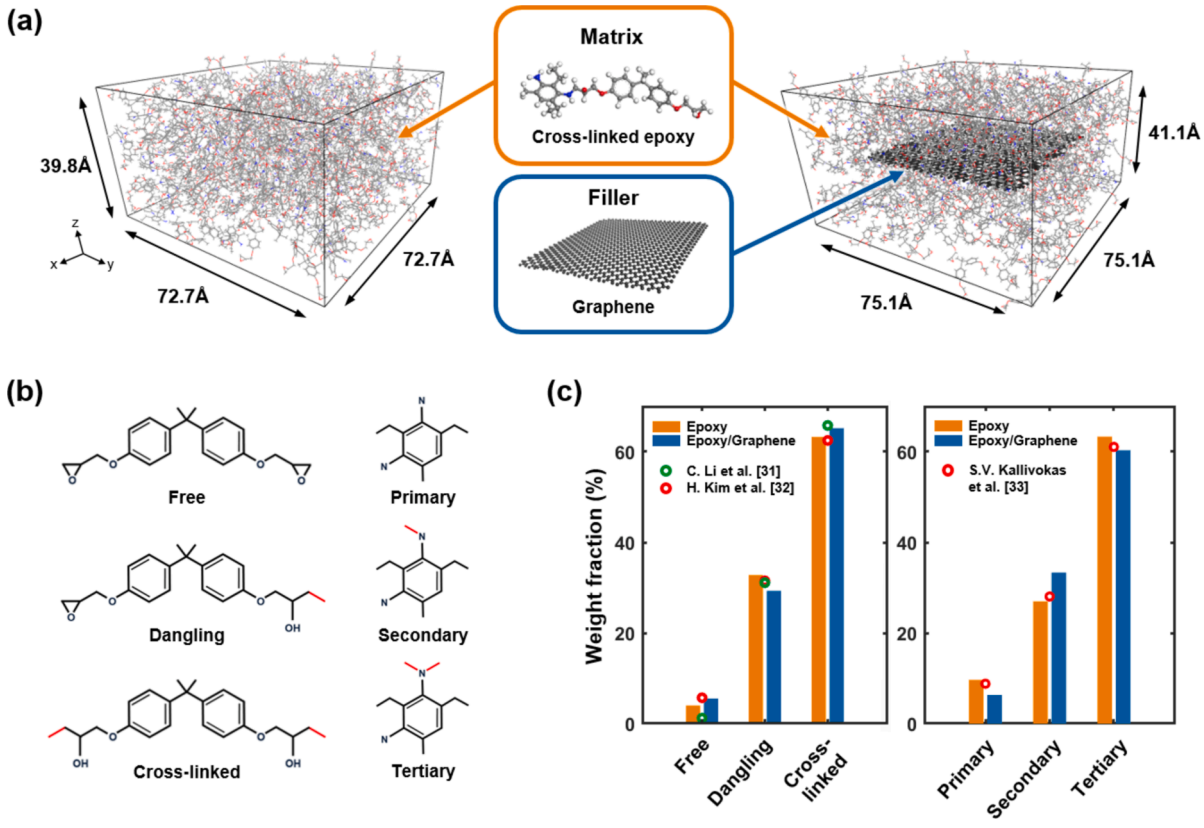



**Fig. 1.** All-atom modeling of cross-linked epoxy functionalized with graphene nano-sheet. (a) Molecular configurations of epoxy and epoxy/graphene nanocomposite models. (b) Classification of epoxy resins and hardeners according to cross-linking state. (c) Weight fraction distribution of resin and hardener by chemical structure.

Under the NPT ensemble at 300 K and 1 atm, the unit cells were deformed at an engineering strain rate of $10^9$ /s until the applied strain reached 12 %. We also show that the considered strain rate is sufficient to produce a converged quasi-static response of the considered microstructure (see **Supplementary Note 2** and **Fig. S2**). During deformation, the atomic stresses in the microstructures were estimated based on the virial theorem, which is defined as:

$$\boldsymbol{\sigma} = \frac{1}{V}\left( -\sum_i^N m_i(\boldsymbol{v}_i \boldsymbol{v}_i^T) + \frac{1}{2}\sum_i^N \sum_{j\neq i}^N \boldsymbol{r}_{ij}\boldsymbol{F}_{ij} \right) \quad (1)$$

where $V$ is the volume, $N$ is the number of constituent atoms, $m_i$ and $\boldsymbol{v}_i$ are the mass and velocity of the $i$-th atom, respectively, and $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ indicate the relative position and interaction force, respectively, between $i$-th and $j$-th atoms [37,38]. To investigate the influence of graphene alignment, the results of the nanocomposite model were subdivided into in-plane and out-of-plane directions. The elastic and shear moduli were calculated from the slope of the stress–strain curve at strains below 3 %.

### 2.3. Oscillatory shear simulation

The damping performance of nanocomposites subjected to oscillatory shear was quantified according to the components that make up the system and their spatial locations. The vibration simulation is conducted by repeatedly applying shear strain to the nanocomposite unit cell. Each plane of the model was subjected to a sinusoidal shear strain as given by the following equation:

$$\gamma(t) = \gamma_0 sin(\omega t), \quad (2)$$

where $\gamma_0$ and $\omega$ are the maximum shear strain and frequency of the oscillations, respectively. The maximum shear strain was varied from 2 % to 10 %. During deformation, the atomic stress in the model was derived based on the virial theory and fitted to a sinusoidal function as follows:

$$\tau(t) = \tau_0 sin(\omega t + \delta), \quad (3)$$

where $\tau_0$ is the maximum amplitude of the stress, and δ is the phase lag between stress and strain profiles.

## 3. Results and discussion

### 3.1. Static mechanical tests of epoxy and epoxy/graphene nanocomposites

The moduli of the epoxy and epoxy/graphene nanocomposites and the corresponding stress–strain curves are shown in Table 1 and Fig. 2, respectively. The tensile modulus of neat epoxy model is 3.02 GPa, which is consistent with previous experimental results (2.9 GPa [39]). The calculated shear modulus was 807.34 MPa, which was in good agreement with earlier simulation reports for the same composition (991.84 MPa [41]). These results confirmed that the force field and equilibration processes in the MD simulation environment effectively reproduced the mechanical properties of the considered materials.

Compared to the neat epoxy model, the graphene-embedded nanocomposite model showed distinct mechanical behavior. However, the tensile properties were not significantly different from those of neat epoxy, which suggests that finite-length graphene flakes do not enhance the physical properties in the longitudinal direction [42,43]. The limited reinforcement of the filler observed in longitudinal loading can be attributed to interfacial slippage between the epoxy and graphene. The small projected area of graphene in the loading direction, combined with slippage effects, hinders effective load transfer to the nanofiller. The similarity between the tensile properties of the graphene nanocomposites and neat epoxy observed here is in good agreement with the experimental observations of Bortz et al. [44]. However, the mechanical response of the graphene nanocomposite model under simple shear deformation was clearly different from that of neat epoxy. There was no significant difference from the neat epoxy in the direction of in-plane deformation, but the mechanical response was significantly reduced for out-of-plane deformation. Specifically, the load in the direction parallel to the interface, where the graphene and epoxy were in contact, did not increase the internal stress of the system, and a significant amount of energy was dissipated. We consider this anisotropic property to be a key factor in improving the energy dissipation and resulting vibration damping properties of the polymer system through graphene insertion [45]. Furthermore, the results suggested that the damping performance of the nanocomposite system was critically related to the alignment direction of the embedded graphene and the loading direction.

### 3.2. Characterization of vibration damping capability by load direction

Results of the static mechanical test suggest that the load transfer and energy dissipation behaviors of the nanocomposites are highly dependent on the insertion and alignment of heterogeneous materials. In particular, the energy dissipation at the interface was found to be most pronounced in the out-of-plane shear loading mode of graphene. Fig. 3 **(a)** shows the process of performing the vibration simulation by repeatedly applying shear strain to the nanocomposite unit cell.

Oscillatory shear deformation was achieved by distorting the simulation box under the NVT ensemble. A shear frequency of $2 \times 10^{10}$ Hz was used, which is of a similar order of magnitude to that achieved in the experimental quartz crystal microbalance method [46]. It is also worth noting that the unit cell equilibrated at 300 K was quenched to 0.1 K for 500 ps before applying shear deformation. This prevented the velocity distribution of each atom within the unit cell from being artificially calibrated by the thermostat under the NVT ensemble [47]. By subjecting nanocomposite microstructures to cryogenic environments, the virial stresses developed by the subsequent mechanical testing can be made to depend purely on the interatomic potential energy. This strategy is particularly useful in quantifying the mechanical contributions of specific regions or components within a simulation box under an ensemble environment [48,49]. The temperature convergence to 0.1 K was completed in a very short time of approximately 30 ps, and the conformations of polymeric molecules equilibrated at room temperature were almost preserved.

Owing to the viscoelastic behavior of the epoxy polymer, a distinct phase difference between the stress and strain profiles was observed during oscillatory shear deformation, as shown in Fig. 3**(b)**. This phase

**Table 1**
Comparison of the structural and tensile properties of the considered models with other literatures.

| | Materials | Approach | Resin/hardener | Cross-linking ratio (%) | Density ($g/cm^3$) | Elastic modulus (GPa) | Shear modulus (MPa) |
|---|---|---|---|---|---|---|---|
| Current study | Epoxy | MD simulation | DGEBA/DETDA | 76.8 | 1.095 | 3.02 | 807.34 |
| | Epoxy/Graphene, in-plane | | | 76.2 | 1.12 | 3.15 | 855.18 |
| | Epoxy/Graphene, out-of-plane | | | | | 3.19 | 519.15 |
| Misumi et al.[39] | Epoxy | Experiment | | – | – | 2.9 | – |
| Rafiee et al. [40] | Epoxy | Experiment | | – | 1.08 | 2.72 | – |
| Clancy et al. [41] | Epoxy | MD simulation | | 80 | 1.08 | 2.75 | 991.84 |

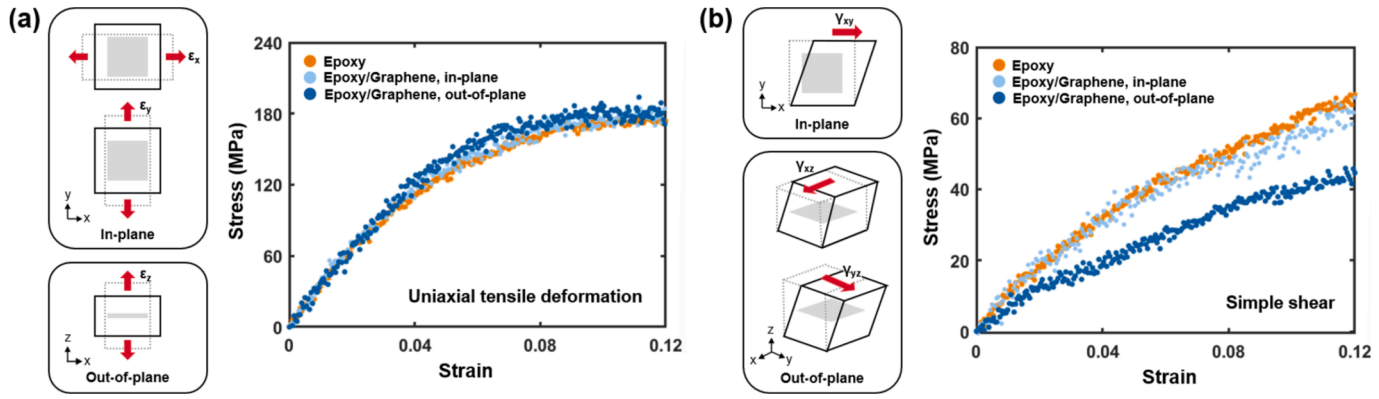



**Fig. 2.** Stress–strain curves obtained from (a) uniaxial tensile tests and (b) simple shear tests.

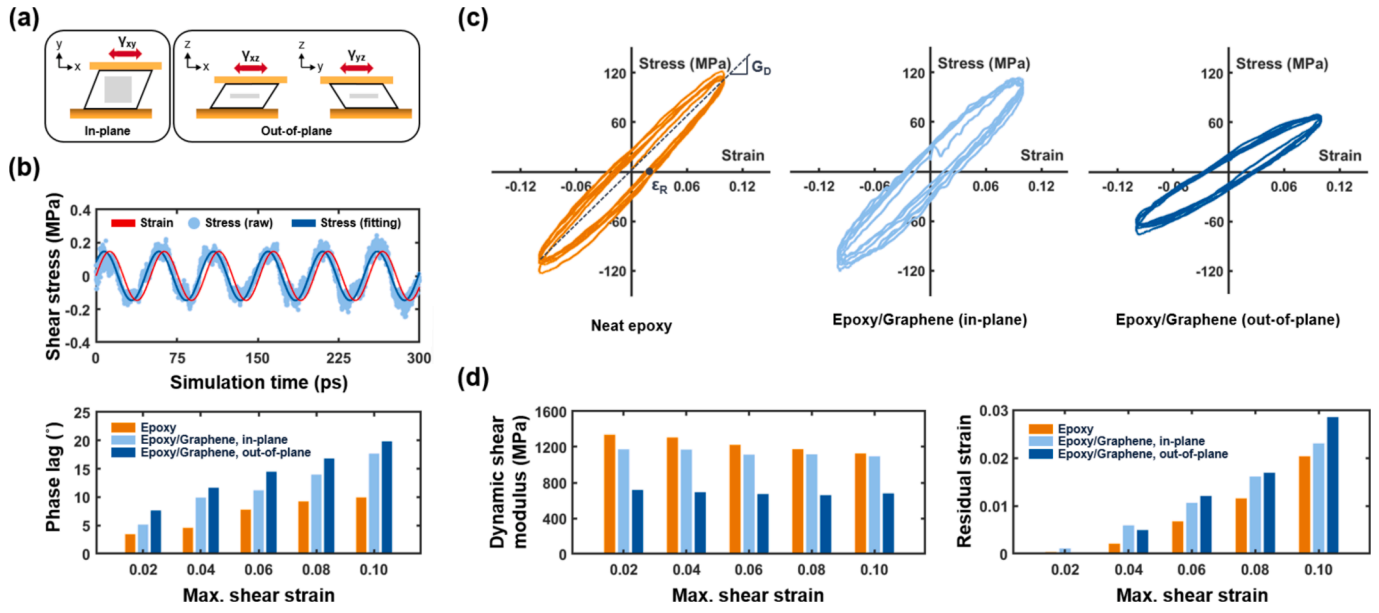



**Fig. 3.** Damping capability of epoxy and epoxy/graphene nanocomposites. (a) Schematic of oscillatory shear tests performed in a simulation environment. The direction of the inserted graphene flake is indicated by a gray square. (b) Representative example of phase lag behavior between strain input and stress response (top) and results organized according to maximum strain (bottom). (c) Stress–strain hysteresis loop during shear oscillation at a maximum strain of 10%. (d) Changes in dynamic shear modulus ($G_D$) and residual strain ($\varepsilon_R$) depending on the applied shear strain.

lag was larger in the presence of graphene than neat epoxy and was especially pronounced under out-of-plane shear loading. This suggests that the enhanced energy dissipation in the nanocomposites was mainly due to imperfect contact and slippage at the interface. Such differences are also clearly visible in the stress–strain hysteresis loops of each model formed during the oscillatory shear simulations, as shown in Fig. 3**(c)**. A distinct closed curve was observed during cyclic loading, attributed to the viscoelastic properties and interfacial friction.

The obtained mechanical behaviour was further analyzed based on two parameters: dynamic shear modulus ($G_D$) and residual strain ($\varepsilon_R$). The dynamic shear modulus was determined by the ratio of the stress range to the strain (i.e., the slope of the major axis of the hysteresis loop) as follows:

$$G_D = |G^*| = \frac{\Delta\tau}{\Delta\gamma} = \frac{\tau_{max} - \tau_{min}}{\gamma_{max} - \gamma_{min}} \tag{4}$$

where $\tau$ and $\gamma$ denote the shear stress and strain, respectively; and $|G^*|$ represents the complex modulus, which characterizes the viscoelastic behavior of polymer materials [50]. The dynamic shear modulus quantifies the load-transfer efficiency under oscillatory loading conditions, with higher values indicating better mechanical resistance to external dynamic loads. As shown in Fig. 3**(d)**, neat epoxy has the largest modulus and smallest values for the out-of-plane shear load of the nanocomposites, which is consistent with the simple shear test results in the previous section. However, materials that dissipate more energy generally have lower dynamic moduli owing to energy losses due to viscoelastic behavior and internal friction [51,52]. From this perspective, the results confirm that the graphene-inserted nanocomposite model exhibits structural anisotropy that becomes softer against out-of-plane loading.

The residual strain quantifies the difference in the system length between the initial and after unloading stages. Owing to the rearrangement and inelastic deformation of the constituent molecules, the length of the entire system along the loading direction increased compared with that at the initial state, even when the stress in the microstructure was completely released. Because internal structural reorganization is a key behavior for energy dissipation in polymer systems, the residual strain under cyclic loading can be used as a parameter to quantify the damping capability of the system. The results show that the residual strain is inversely proportional to the static or dynamic shear modulus and is most pronounced for the out-of-plane shear deformation of epoxy nanocomposites.

First, the individual responses of the nanocomposite components were evaluated. During the shear oscillation, the virial stress tensors of the epoxy matrix and graphene were derived separately using Eq. (1). In the virial stress calculation, the stress tensor is determined for each constituent atom so that the stresses applied to the epoxy and graphene in a single nanocomposite system can be obtained separately. Here, $V$ is substituted by the volume of each component obtained through Voronoi tessellation. Therefore, the stress in the entire nanocomposite is mathematically equivalent to the volume-weighted average of the stresses applied to the epoxy and graphene [53].

The evaluation results of the dynamic shear modulus of the nanocomposite components are shown in Fig. 4**(a)**. The nanocomposites containing graphene nanofillers exhibited a very low apparent shear modulus (<101.32 MPa) regardless of the applied strain. This shows that the region between the graphene and the adjacent epoxy matrix slips easily under shear deformation, resulting in minimal load transfer. From the perspective of the epoxy matrix, the presence of graphene was similar to that of planar voids. Therefore, the dynamic shear modulus of the epoxy matrix is only 0.56 times compared to the neat material. The effect of modulating the mechanical properties of the interphase induced by graphene was confirmed by the change in the dynamic shear modulus according to the relative distance from graphene. The dynamic shear modulus of the epoxy matrix had the lowest value in the region in contact with graphene and then increased with increasing distance.

The dependence of the residual strain formed in the interphase was also interesting. As shown in Fig. 4**(b)**, the average residual strains of both components—bulk epoxy and graphene—exceed those of the neat epoxy by a factor of 1.94 and 1.87, respectively. In particular, when the maximum strain of the externally applied vibration exceeded 6 %, the residual strain of graphene almost converged to a specific value. This contrasts with the results in which the residual strain of the epoxy matrix increased linearly with the applied shear strain. Accordingly, the magnitudes of the residual strains of the two components were reversed under the vibration conditions, with a maximum strain of 8 %. Such differences were also confirmed in the results showing the residual strain of the epoxy matrix as a function of the distance from the graphene. Within the applied strain conditions considered, the residual strain in the epoxy matrix was the lowest in the region in direct contact with graphene and then tended to increase gradually with the distance from graphene. In conclusion, the interphase matrix adjacent to graphene is characterized by low effective stiffness as well as low energy dissipation efficiency under oscillation shear deformations. The phase lag was also used to quantify the viscoelastic properties of the interphase and bulk regions, which has a consistent result with the current study (see **Supplementary Note 1** and **Fig. S1**).

### 3.3. *Interphase network structure modified by graphene and its influence on damping capacity*

The findings discussed in Section 3.2 indicate that the region surrounding graphene exhibits mechanical behavior significantly different from that of the bulk matrix [54]. These observations are consistent with previous computational studies that have quantitatively discussed the spatial deformation of polymer matrices influenced by interaction forces of nanofillers [55]. Specifically, the vibration characteristics of the epoxy that constitutes the nanocomposite system must be understood with regional specificity, considering the morphological modifications induced by the interaction with graphene. Thus, the structural reconfiguration of the polymer networks in proximity to graphene was investigated in greater detail.

Upon interaction with a heterogeneous filler, such as graphene, the polymer matrix underwent four principal structural alterations (Fig. 5 **(a)**). First, a densely packed polymer layer was formed adjacent to the graphene interface because of the strong adsorption forces exerted by the filler. The system was then divided into multiple cubic shells of 0.2 Å thickness, and the density of each shell was measured (Fig. 5**(b)**). The density profile showed a peak of 1.61 g/cm$^3$ at a distance of 3.6 Å from the graphene sheet. As the distance was further increased, the density profile gradually converged to the bulk value of 1.12 g/cm$^3$ [56]. Based on this observation, we defined the boundary between the bulk and interphase as the point at which the density first converged to that of neat epoxy (6.6 Å). Second, the polymer layer adsorbed by the

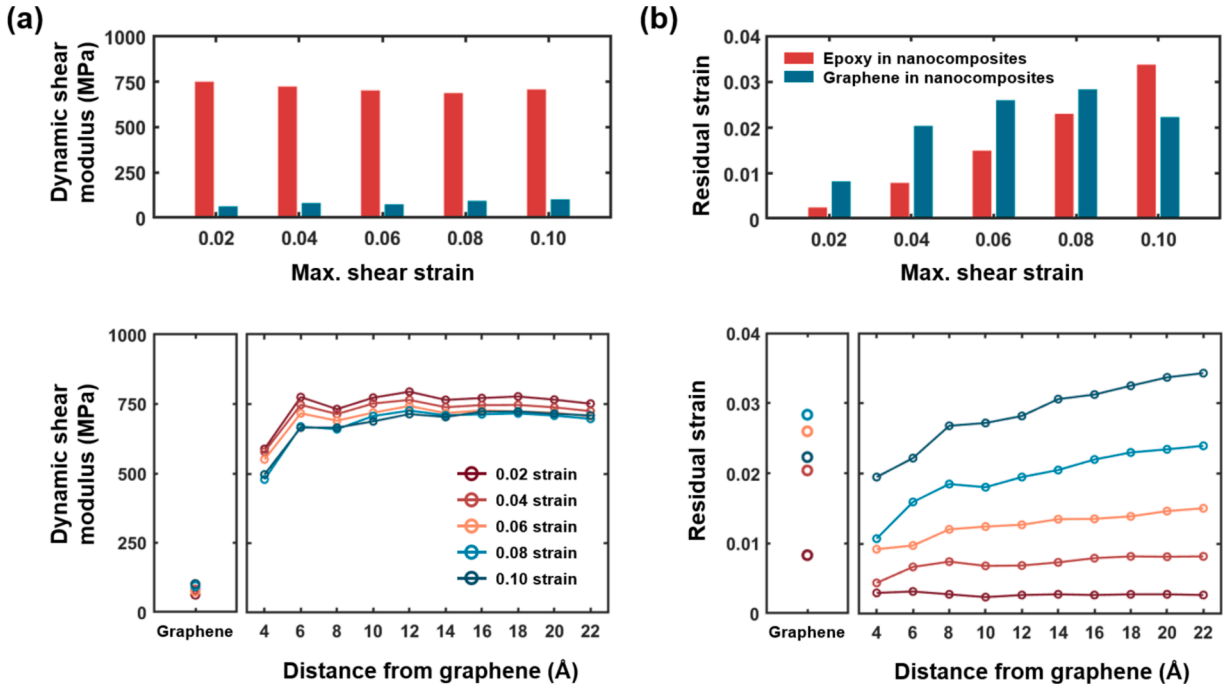



**Fig. 4.** Evaluation of the stiffness and damping properties of individual components within the nanocomposites. (a) Dynamic shear moduli of nanocomposites subjected to out-of-plane shear strain (top) and their spatial distributions based on distance from the graphene (bottom). (b) Distributions of residual strains by component (top) and by spatial domain (bottom) in the same environment.

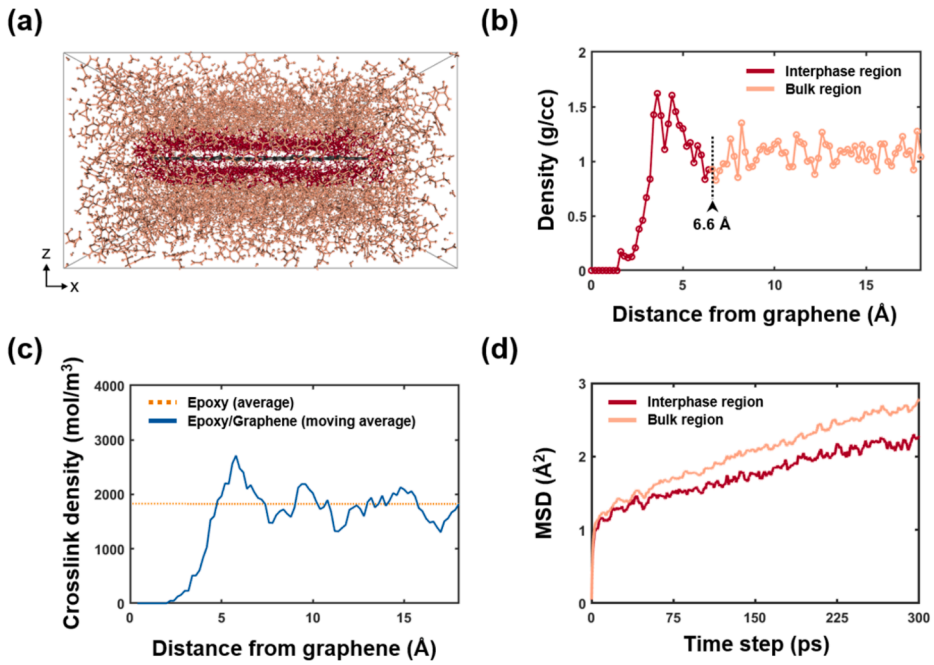



**Fig. 5.** Distinct characteristics of the interphase and bulk region of epoxy. (a) Configuration of the epoxy/graphene unit cell depicting the interphase and bulk regions. (b) Density distribution of epoxy matrix around the graphene surface. (c) Distribution of crosslinking density in epoxy region according to distance from graphene. To reduce data scattering and improve visibility, the profiles of epoxy/graphene nanocomposites were represented by calculating the moving average of values over a region of 2 Å thickness. (d) Mean square displacement profiles of atoms in the interphase and bulk regions under thermodynamic equilibrium conditions.

heterogeneous filler exhibited a locally high cross-linking density. When the distribution of the cross-linking density was obtained from the surface in the same manner (Fig. 5**(c)**), the cross-linking density profile showed a significant correlation with the mass density distribution. This was attributed to the localized aggregation of the molecular components, which enhanced the proximity of the cross-linking sites. Third, the polymers captured in heterogeneous fillers exhibited retarded chain dynamics. This phenomenon was quantified by the mean square displacement (MSD) analysis of the bulk region and the interphase (Fig. 5**(d)**). Clearly, the bulk region exhibited a much steeper increase in the MSD curve than the area near the filler. This is primarily due to the anchoring effect of graphene, which restricts the chain dynamics at the interphase. Finally, graphene behaved as a spatial obstacle, impeding the formation of a continuous three-dimensional network. The van der Waals gap between the epoxy matrix and graphene effectively inhibits molecular interpenetration, as evidenced by the absence of cross-linking density at distances of ~ 2 Å from the graphene. Consequently, the epoxy matrix represents a network structure that bypasses graphene.

In summary, the morphology of the graphene-modified epoxy interphase is characterized by a region of high cross-linking density, which enhances the local elasticity. The interphase is understood to be a network structure surrounding a planar-shaped nanopore with a very dense, highly cross-linked epoxy. The layered structural nature of such an interphase explains the relatively low load-transfer efficiency between graphene and epoxy, as observed in Fig. 2**(b)**. It should be highlighted that the planar pore model of the graphene sheet is specific to out-of-plane shear deformations. In this mode, the sliding displacement between epoxy and graphene plays a critical role. The epoxy component facilitates the rupture and regeneration of π-interactions due to the repetitive hexagonal structure of graphene, enabling the system to reach its lowest energy state following displacement and promoting easier interfacial slip. In contrast, the planar pore model of graphene sheet must be distinguished from the vertical delamination mode observed under out-of-plane uniaxial tension, which exhibits no degradation in mechanical properties (Fig. 2), as the graphene maintains cohesive interfacial interactions with the epoxy matrix (see Fig. S3).

The deformation behavior of the interphase and bulk regions was further characterized using atomistic local strain. The separation vector d of atom $i$ for its neighboring atom $j$ is defined as follows:

$$\boldsymbol{d}_{ji} \equiv \boldsymbol{x}_j - \boldsymbol{x}_i \tag{5}$$

The separation vector $\boldsymbol{d}$ is time-dependent. The original separation is given by:

$$\boldsymbol{d}^0_{ji} \equiv \boldsymbol{x}^0_j - \boldsymbol{x}^0_i \tag{6}$$

A locally affine transformation matrix $\boldsymbol{J}_i$ is derived from the following equation:

$$\sum_{j\in N_i}\left|\boldsymbol{d}^0_{ji}\boldsymbol{J}_i - \boldsymbol{d}_{ji}\right|^2 = \sum_{j\in N_i}\left(\boldsymbol{d}^0_{ji}\boldsymbol{J}_i - \boldsymbol{d}_{ji}\right)\left(\boldsymbol{d}^0_{ji}\boldsymbol{J}_i - \boldsymbol{d}_{ji}\right)^T \tag{7}$$

$$= \mathrm{Tr}\sum_{j\in N_i}\left(\boldsymbol{J}_i^T\boldsymbol{d}^{0T}_{ji} - \boldsymbol{d}^T_{ji}\right)\left(\boldsymbol{d}^0_{ji}\boldsymbol{J}_i - \boldsymbol{d}_{ji}\right) \tag{8}$$

where the integer $N_i$ is the number of neighbors of atom $i$ in the present configuration. Using the affine transformation matrix, the Lagrangian strain tensor $\boldsymbol{\varepsilon}_{ji}$ is calculated as:

$$\boldsymbol{\varepsilon}_{ji} = \frac{1}{2}\left(\boldsymbol{J}_i\boldsymbol{J}_i^T - \boldsymbol{I}\right). \tag{9}$$

The strain tensor for each atom within 15 Å is obtained individually and averaged. The von Mises strain $\varepsilon_{vM}$ is calculated as:

$$\varepsilon_{vM} = \sqrt{\varepsilon^2_{xy} + \varepsilon^2_{yz} + \varepsilon^2_{xz} + \frac{\left(\varepsilon_{xx} - \varepsilon_{yy}\right)^2 + \left(\varepsilon_{yy} - \varepsilon_{zz}\right)^2 + \left(\varepsilon_{zz} - \varepsilon_{xx}\right)^2}{6}} \tag{10}$$

$$= \frac{\sqrt{2}}{3}\sqrt{\left(\varepsilon_1 - \varepsilon_2\right)^2 + \left(\varepsilon_2 - \varepsilon_3\right)^2 + \left(\varepsilon_3 - \varepsilon_1\right)^2} \tag{11}$$

where $\varepsilon_1$, $\varepsilon_2$, and $\varepsilon_3$ are the principal strains.

The strain distribution in Fig. 6 was calculated using atomic coordinates immediately after equilibration as the reference state,

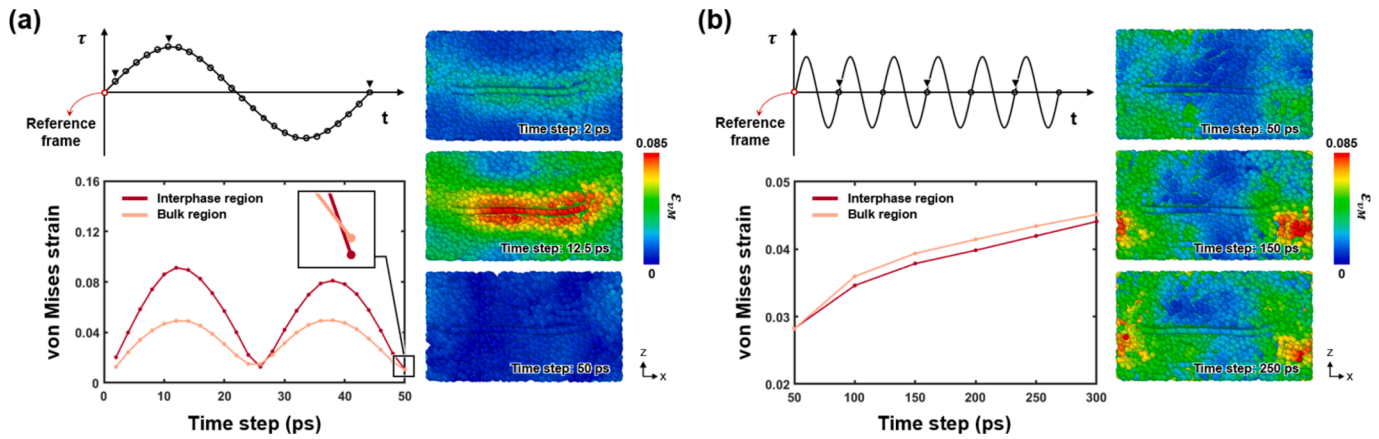



**Fig. 6.** Von Mises strain analysis using the finite strain theory. (a) Strain distributions in the interphase and bulk regions during single and (b) multiple cycles. Each figure presents the applied shear strain over simulation time (top left), the resulting von Mises strain profiles (bottom left), and a visualization of the atomic strain distribution (right). In the figure of applied shear strain, the black circular markers indicate the frame in which the von Mises strain was measured, and the red circular marker represents the reference frame. The frame from which the snapshot was taken is highlighted as an inverted triangle (▼).

representing changes during the mechanical test. Graphene atoms were included in the von Mises strain calculation for the interphase to prevent inaccuracies that could arise from treating the interphase as a free surface. Including graphene atoms in the strain calculations offers the advantage of directly visualizing the relative displacements between the graphene and epoxy matrix. This approach enables a reliable evaluation of local slip, reorientation, and mechanical response at the interface during deformation of the entire system. The von Mises strain distribution during a single shear-deformation cycle is shown in Fig. 6**(a)**. The mechanical responses of the interphase and bulk regions are clearly distinguishable. During shear deformation, the von Mises strain accumulated more intensively in the interphase than in the bulk. This is plausible considering the unstable and imperfect contact between epoxy and graphene, that is, the externally applied shear strain promotes immediate deformation in the interphase. The inserted graphene acted as nano-sized voids within the nanocomposites and disrupted the formation of a continuous cross-linked network, causing the epoxy near the interface to experience amplified strain. Interestingly, the residual von Mises strain after each cycle was more pronounced in the bulk matrix. This suggests that the interphase network exhibited excellent structural recovery ability. The results presented in Fig. 5**(c)** clearly show that the polymer layer adsorbed on the graphene surface led to a locally higher cross-linking density. The increased cross-linking density prevents the microscopic rearrangement of network components in the area where the interphase and bulk are in direct contact, imparting higher structural elasticity.

As shown in Fig. 6**(b)**, such domain-specific distinct characteristics of the microstructure were similarly observed in the von Mises strain analysis over multiple cycles. In this environment, a linear accumulation of the von Mises strain with an increasing number of deformation cycles was also observed. Notably, the accumulation of this strain was always lower in the interphase than in the bulk region. The molecular origin of vibrational absorption in polymer systems is the microscopic redistribution of molecular components, which was captured by higher von Mises strains in the present approach. Therefore, this behavior supports the finding that the epoxy has a higher dynamic shear modulus and residual strain with increasing distance from graphene.

Based on the observations made thus far, the general interphase model proposed in this study is as follows: The simplified cross-sectional structure of the epoxy nanocomposites functionalized with graphene nanofillers is shown in Fig. 7**(a)**. The material heterogeneity between graphene and epoxy induces two major structural changes. First, because graphene adsorbs the surrounding epoxy components on its surface, the dense matrix region under the interaction effect of the filler exhibits a locally high cross-linking density. Second, a distinct physical gap of about 2 Å is created around the graphene surface, corresponding to the equilibrium distance of non-covalent bonds between the graphene and the polymer chains. The deformation characteristics of

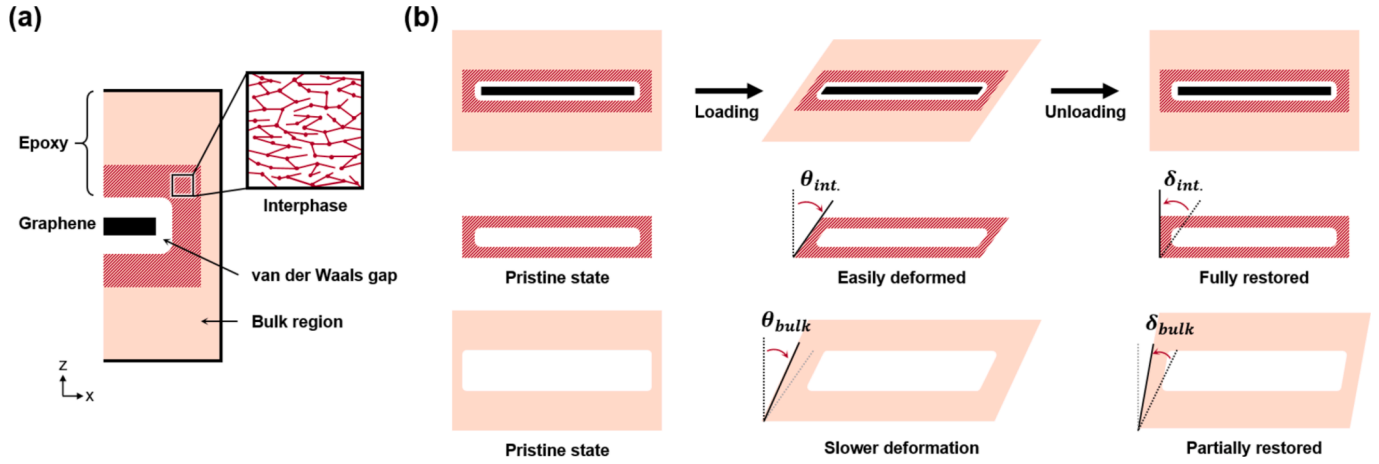



**Fig. 7.** Schematic illustration of the distinct responses of the bulk and interphase matrices. (a) Simplified cross-sectional model of the epoxy/graphene nanocomposite and (b) structural behavior of individual domains during dynamic shear oscillation.

nanocomposites with such an interphase structure when shear strain is applied are presented in Fig. 7**(b)**. When a perfect bonding is not achieved between epoxy and graphene, the matrix behaves as if it has planar pores inside, causing more severe deformations to occur at the interphase ($\theta_{int.} > \theta_{bulk}$). In contrast, areas with locally higher cross-linking densities had stronger elasticity. Therefore, during subsequent unloading, the interphase network quickly recovers its initial structure, while significant residual deformation remains in the far bulk region ($\delta_{int.} > \delta_{bulk}$).

The findings of this study suggest that the vibration damping performance of epoxy/graphene systems can be further enhanced by refining the interphase design. For example, strategies that intentionally reduce the cross-linking density in the interphase can lead to beneficial vibrational properties. Pre-coating graphene with organic materials, including dopamine, ellipticine, and pyrenebutyric acid [57–59], can suppress the formation of covalent bonds in the interphase network by increasing the distance between the matrix components. This approach leads to improved intermolecular friction under shear loading by lowering the elasticity of the interphase network, as desired by the user. The application of planar-structured hardeners in the interphase is another promising approach. Planar-structured hardeners achieve stronger adsorption to graphene than to the resin during the synthesis step because of their structural affinity, allowing for spatial variation in the resin-to-hardener ratio. Therefore, by designing the interphase with excessive hardener content in this manner, a relatively sparsely cross-linked interphase could be achieved during the subsequent curing process.

## 4. Conclusion

In this study, we obtained theoretical insights into the damping properties of graphene-containing epoxy nanocomposites using all-atom MD simulations. The anisotropy due to the difference between the direction of the applied shear vibration and the orientation of graphene was numerically characterized. In particular, the behavior of the epoxy network structure in the graphene-modified interphase was clearly distinguished from that of the bulk. These structural properties were analyzed in relation to the mechanical behaviors, such as the dynamic shear modulus and residual strain, which were applied differently to each component and region of the nanocomposites.

According to the results, the structural changes in the epoxy matrix because of graphene insertion can be attributed to the following four characteristics: (1) A densely packed layer was created owing to the strong adsorption forces exerted by the filler. (2) The polymer layer absorbed by the heterogeneous filler exhibits a locally high cross-linking density. (3) Polymers captured in heterogeneous fillers exhibited retarded chain dynamics. (4) Graphene acted as a spatial obstacle that impeded the formation of continuous three-dimensional network. The epoxy region in direct contact with graphene exhibited a small residual strain despite its low elastic modulus. This is a unique property of epoxy/graphene nanocomposites and is distinct from the trade-off between the damping and stiffness observed in homogeneous materials.

The mechanism presented in this study also suggests that the shear strength properties of nanocomposites are significantly impaired under cyclic shear loading compared with those of neat materials, particularly when shear strain is applied in the direction parallel to the graphene insertion direction. The incomplete bond formed at the epoxy/graphene interface region causes large friction because of external deformation, which results in high energy dissipation during the process. Changes in the vibrational characteristics are observed not only in the interphase, where the cross-linking density and structural density are changed by graphene, but also throughout the entire region occupied by the epoxy matrix. Specifically, the damping capacity of the epoxy matrix within the nanocomposites is spatially dependent and exhibits distinct behavioral differences depending on the relative distance from graphene.

The findings of this study pave the way for the microstructural designing of epoxy/graphene systems used for vibration damping. The changes expected in energy dissipation characteristics according to the varying composition and spatial domain of each material are explained from a nanostructural perspective. Therefore, this study serves as a theoretical guideline for the rational microstructural design of graphene-embedded nanocomposites. Furthermore, the methodology developed in this study offers a quantitative framework for analyzing the spatial distribution of damping capabilities in nanocomposites. In the near future, the methodology will be further expanded to explore the molecular mechanisms of simultaneously changing the behavior of the entire nanocomposite and the morphological properties of the interface, such as physical/chemical modification of the nanofiller surface.

## CRediT authorship contribution statement

**Sihyun Kim:** Writing – original draft, Visualization, Software, Methodology, Data curation. **Hongdeok Kim:** Writing – review & editing, Visualization, Software, Methodology. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This study was supported by a National Research Foundation of Korea (NRF) grant (RS-2023-00210865) funded by the Korean government (MSIT).

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compositesa.2025.108720.

## Data availability

Data will be made available on request.

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